

Real-Time Malware Detection and Analysis using Wazuh and VirusTotal Integration

Project Overview

Modern endpoint security requires **real-time detection, enrichment, and contextual analysis** of potentially malicious files. Traditional File Integrity Monitoring (FIM) alone can detect file changes, but it cannot determine whether a file is actually malicious.

This project demonstrates how **Wazuh**, an open-source SIEM and XDR platform, can be integrated with **VirusTotal** to automatically analyze newly created or modified executable files in real time. By combining **Wazuh FIM** with **VirusTotal threat intelligence**, security teams gain immediate visibility into malware presence, severity, and threat context.

The project focuses on:

- Detecting executable file creation/modification
- Submitting file hashes to VirusTotal
- Reducing alert noise and API overuse
- Enriching alerts with VirusTotal verdicts
- Mapping detections to the MITRE ATT&CK framework
- Visualizing alerts in Wazuh Dashboard and TheHive



Why Integrate Wazuh with VirusTotal?

CHALLENGE	SOLUTION
FIM DETECTS FILE CHANGES BUT NOT MALICIOUS INTENT	VirusTotal provides malware verdicts
HIGH ALERT NOISE FROM FILE CHANGES	Custom rules suppress clean results
LACK OF THREAT CONTEXT	VirusTotal adds reputation, engines, links
MANUAL MALWARE INVESTIGATION	Automated enrichment
NO THREAT INTELLIGENCE CORRELATION	MITRE mapping include

Tools & Technologies Used

- Wazuh Manager (SIEM / XDR)
- Wazuh Agent (Ubuntu 24.04)
- VirusTotal Public API
- Wazuh Dashboard
- TheHive (Case Management)
- File Integrity Monitoring (FIM)
- MITRE ATT&CK Framework

Architecture Overview

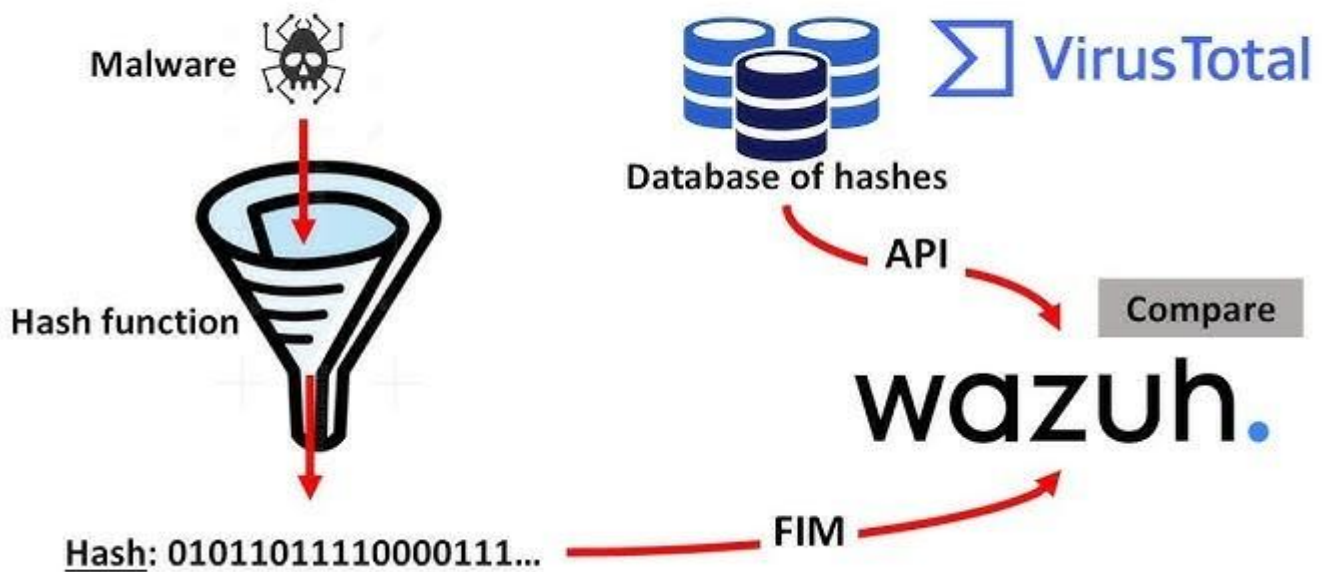
Ubuntu Endpoint

- └─ Wazuh Agent
 - └─ FIM monitors directory (real-time)
 - └─ Detects executable file change
 - └─ Sends alert to Wazuh Manager

Wazuh Manager

- └─ Custom Rules (noise control)
- └─ VirusTotal Integration
- └─ Alert Enrichment
- └─ MITRE Mapping
- └─ Dashboard & TheHive Visualization

How Wazuh Integrates with VirusTotal



Step-by-Step Implementation

Step 1: VirusTotal Account Setup

- Created a VirusTotal account
- Generated a **public API key**
- Noted API limitations:
 - 4 requests/min
 - 500 requests/day
- This influenced alert and rule design to **control noise and rate limits**

The screenshot shows the VirusTotal API Key page. At the top, there is a search bar and a user profile. Below this, the 'API KEY' section displays a warning about the key's confidentiality and a link to the Terms of Service. The 'API QUOTA ALLOWANCES FOR YOUR USER' section shows the user's current status as a standard free end-user account with the following limitations:

Access level	Usage	Request rate	Daily quota	Monthly quota
△ Limited, standard free public API	Must not be used in business workflows, commercial products or services.	4 lookups / min	500 lookups / day	15.5 K lookups / month

Below the table, there are links to 'Go premium', 'Use in browser', 'Discover feeds', and 'Other services'. The page also includes a link to 'Upgrade to premium' and a link to 'Upgrade API'.

Step 2: Enable VirusTotal Integration on Wazuh Server

Edited the Wazuh manager configuration file:

File:

/var/ossec/etc/ossec.conf Added:

```
<integration>
  <name>virustotal</name>
  <api_key>...../api_key> <!-- Replace with your VirusTotal API key -->
  <rule_id>100027,100028</rule_id>
  <alert_format>json</alert_format>
</integration>
```

Purpose:

- Triggers VirusTotal scans **only** when specific rules fire
- Prevents scanning every file change
- Keeps API usage under control

Step 3: Custom Rules to Control Noise and API Usage

Created custom rules to:

- Allow only **executable files**
- Suppress clean VirusTotal results
- Raise high-severity alerts for real malware

File:

/var/ossec/etc/rules/local_rules.xml **Key**

Rules Implemented:

```
<group name="syscheck,malware,virustotal">

  <!-- New executable file -->
  <rule id="100027" level="12">
    <if_sid>550</if_sid>
    <match>\.(exe|elf|sh|py|js|ps1|bat)$</match>
    <description>Executable file added - allowed for VirusTotal scan</description>
  </rule>

  <!-- Modified executable file -->
  <rule id="100028" level="12">
    <if_sid>554</if_sid>
    <match>\.(exe|elf|sh|py|js|ps1|bat)$</match>
    <description>Executable file modified - allowed for VirusTotal scan</description>
  </rule>

  <!-- Suppress clean VirusTotal results -->
  <rule id="100029" level="0">
    <if_sid>87104</if_sid>
    <field name="data.virustotal.positives">^0$</field>
    <description>VirusTotal clean result - suppressed</description>
    <options>no_log</options>
  </rule>

  <!-- High confidence malware: create TheHive case -->
  <rule id="100030" level="15">
    <if_sid>87105</if_sid>
    <field name="data.virustotal.positives">^[5-9]|[1-9][0-9]+$</field>
    <description>High confidence malware detected by VirusTotal</description>
    <mitre>T1059</mitre>
  </rule>

</group>
```

Why This Matters:

- Prevents unnecessary VirusTotal lookups
- Avoids alert fatigue
- Highlights only **real threats**

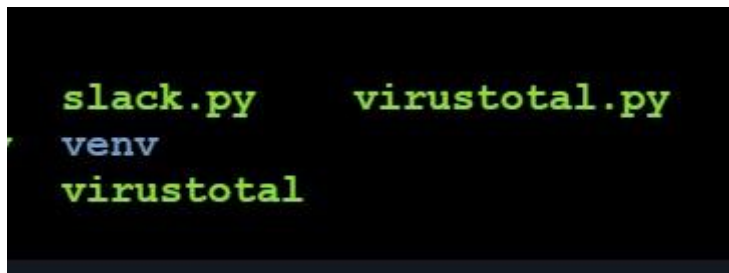
Step 4: Verify VirusTotal Integration Script

Confirmed the default VirusTotal integration script exists:

Directory:

/var/ossec/integrations/

- virustotal.py
- slack.py (optional)
- These scripts are **prebuilt by Wazuh**
- No modification required



Step 5: Configure FIM on Wazuh Agent (Ubuntu Endpoint)

Edited agent configuration:

File:

/var/ossec/etc/ossec.conf

Added monitored directory:

```
<syscheck>
  <directories check_all="yes" realtime="yes">/media/user/software</directories>
</syscheck>
```

Important:

- VirusTotal scans are triggered **only for this directory**
- Files in other directories will **NOT** trigger VirusTotal

Step 6: Malware Test on Ubuntu Endpoint

Downloaded EICAR test malware:

```
root@ip-10-10-10-10:~# sudo curl -Lo /media/user/software/suspicious-file.exe https://secure.eicar.org/eicar.com
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload  Total   Spent    Left     Speed
100    68  100    68    0     0    173      0  0:00:00  0:00:00  0:00:00  173
```

This triggered:

- FIM detection
- Custom executable rule
- VirusTotal hash lookup

Step 7: Wazuh Dashboard Alert Analysis

Observed alerts in Wazuh Dashboard:

- File creation detected
- Executable classification
- VirusTotal verdict (67 engines detected)
- MITRE ATT&CK mapping: **Execution**
- File path, hash, permissions, owner

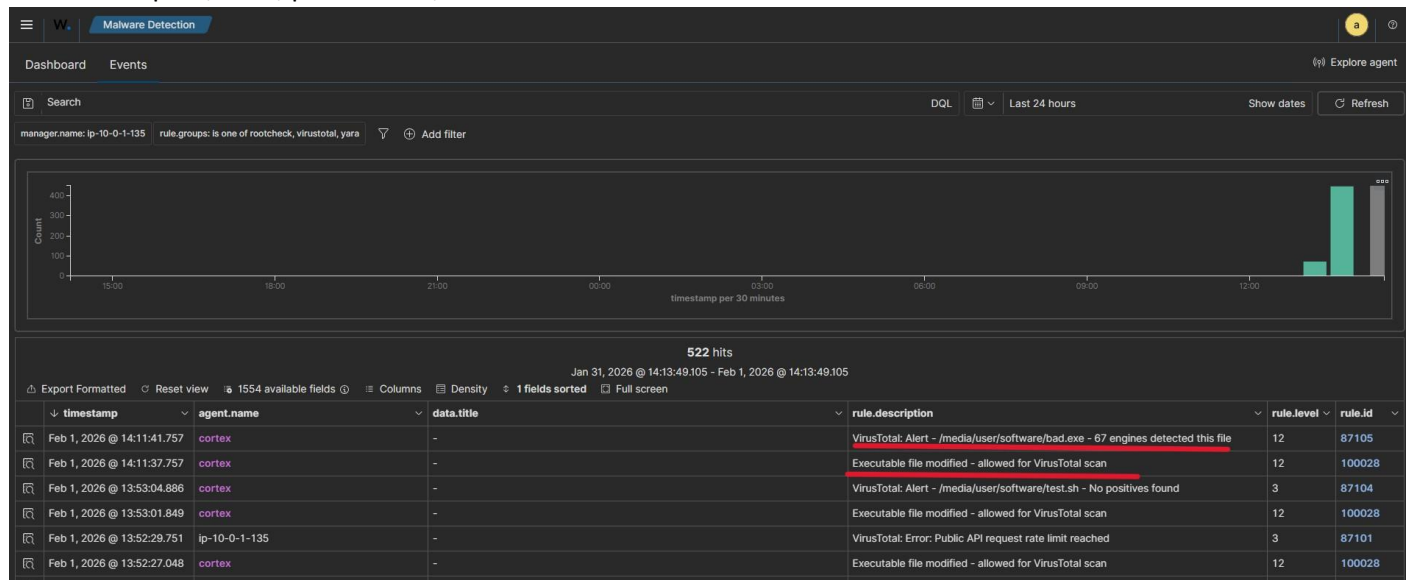


Table JSON

t	_index	wazuh-alerts-4.x-2026.02.01
t	agent.id	005
t	agent.ip	
t	agent.name	cortex
t	decoder.name	syscheck_new_entry
t	full_log	File '/media/user/software/bad.exe' added Mode: realtime
t	id	1769935297.2171281
t	input.type	log
t	location	syscheck
t	manager.name	ip-10-0-1-135
t	rule.description	Executable file modified - allowed for VirusTotal scan
#	rule.firedtimes	1
t	rule.groups	syscheck, malware, virustotal
t	rule.id	100028
#	rule.level	12
🔊	rule.mail	true
t	syscheck.event	added
t	syscheck.gid_after	0
t	syscheck.gname_after	root
t	syscheck.inode_after	542280
t	syscheck.md5_after	44d88612fea8a8f36de82e1278abb02f
t	syscheck.mode	realtime
📅	syscheck.mtime_after	Feb 1, 2026 @ 14:11:37.000
t	syscheck.path	/media/user/software/bad.exe
t	syscheck.perm_after	rw-r--r--
t	syscheck.sha1_after	3395856ce81f2b7382dee72602f798b642f14140
t	syscheck.sha256_after	275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f
#	syscheck.size_after	68
t	syscheck.uid_after	0
t	syscheck.uname_after	root
📅	timestamp	Feb 1, 2026 @ 14:11:37.757

Table	JSON
_index	wazuh-alerts-4.x-2026.02.01
agent.id	005
agent.ip	
agent.name	cortex
data.integration	virustotal
data.virustotal.found	1
data.virustotal.malicious	1
data.virustotal.permalink	https://www.virustotal.com/gui/file/275a021bbfb6489e54d471899f7db9f1663fc695ec2fe2a2c4538aabf651fd0f/detection/f-275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f-1769935144
data.virustotal.positives	67
data.virustotal.scan_date	2026-02-01 08:39:04
data.virustotal.sha1	3395856ce81f2b7382dee72602f798b642f14140
data.virustotal.source.alert_id	1769935297.2171281
data.virustotal.source.file	/media/user/software/bad.exe
data.virustotal.source.md5	44d88612fea8a8f36de82e1278abb02f
data.virustotal.source.sha1	3395856ce81f2b7382dee72602f798b642f14140
data.virustotal.total	69
decoder.name	json
id	1769935301.2172093
input.type	log
location	virustotal
manager.name	
rule.description	VirusTotal: Alert - /media/user/software/bad.exe - 67 engines detected this file
rule.firedtimes	1
rule.gdpr	IV_35.7.d
rule.groups	virustotal
rule.id	87105
rule.level	12
rule.mail	true
rule.mitre.id	T1203
rule.mitre.tactic	Execution
rule.mitre.technique	Exploitation for Client Execution
rule.pci_dss	10.6.1, 11.4
timestamp	Feb 1, 2026 @ 14:11:41.757

Step 8: VirusTotal Threat Intelligence Validation

Clicked VirusTotal permalink from alert:

- Verified detection score
- Confirmed malware classification (EICAR test)
- Viewed detection engines and tags

Project Outcome

Real-time executable detection
VirusTotal threat intelligence enrichment
Noise reduction using custom rules
MITRE ATT&CK mapping
Dashboard visualization
SOC-ready alerts in TheHive

Skills Gained

- Wazuh SIEM configuration
- File Integrity Monitoring (FIM)
- Threat intelligence integration
- Custom rule writing
- Alert noise reduction
- API rate-limit handling
- SOC alert analysis
- MITRE ATT&CK mapping
- End-to-end malware detection workflow

Conclusion

This project demonstrates a **practical SOC-grade malware detection pipeline** using Wazuh and VirusTotal. By carefully controlling which files are scanned and suppressing benign results, the solution remains efficient even with a free VirusTotal API key.

The implementation is scalable, production-ready, and forms a strong foundation for **automated response**, which is addressed separately in the Upcoming Project, **Project 2: Active Response**.
